Cassava Biopolymer Project

Materials Innovation Hub — Club Activity Plan

Project Overview

- Raise campus awareness about plastics and microplastics through content
- Design and synthesize a biopolymer alternative from cassava starch
- Test mechanical, thermal, and degradation properties
- Present findings to stakeholders and submit for publication/ poster presentation or a report memo

The Four Teams

We will divide members into four specialized teams that make sure the all the members follows the specific team objectives and how to achieve it.

Team 1: Awareness & Content

- Creating social media content (Instagram, TikTok) about the plastics crisis, their impact and the project
- Writing a 2 to 3minute explainer video script on why we need to innovate an alternative to plastics
- Documenting the project journey with photos/video for final presentation

Team 2: Research & Formulation

- Literature review: Application of cassava starch in material science engineering as an alternative to plastics (polymer science), cassava starch properties, biopolymer synthesis routes, plasticizers (glycerol)
- Designing 3 to 5 approach variations (different plasticizer types/amounts, temperatures, drying times)
- Researching degradation mechanisms (microbial, thermal, aqueous)
- Creating a testing protocol document (which tests, in what order, success criteria)

Team 3: Polymer Synthesis

- Procuring cassava or extract starch from cassava root
- Mixing formulations according to Team 2s approach
- Heat, cast into films, and cure samples
- Optimizing variables (temperature ramps, curing time, mold prep)
- Documenting process with photos and batch logs

Team 4: Testing & Characterization

- Mechanical testing: tensile strength, elongation, hardness (using available equipment or simple manual tests)
- Thermal analysis: water absorption, thermal stability (TGA/DSC, oven tests)
- Biodegradability tests: soil burial and different weather kinetics, aqueous degradation, composting behavior
- Data analysis and visualization: charts, statistical summary, comparison tables
- Writing the results & discussion section for the final report

Weekly Meeting Structure

Every Saturday 4–6pm.

Time	Duration	What Happens
4:00–4:05	5 min	Quick recap: what did we learn last time? Where are we now?
4:05–4:20	15 min	Team briefs (2 min each).
		Each team: what are we doing today, what do we need, any setbacks?
4:20–5:55	95 min	Building. Teams work independently on their tasks. I will rotate between teams, unblock problems, and capture photos/videos for social media.
5:55–6:00	5 min	Debrief. Share accomplishments. Announce next week's focus.